# Overcoming the Cold-Start Calibration Barrier in Distributed Automotive Radar Re-Simulation via Dual-Horizon Dynamic Alignment and Preflight Seeding

*Paper 1 of 2 | A-PRISM programme | target venue: IEEE, November submission*

## I. The 16-Log Experiment and the Wall We Hit

The problem that started this was not a research question. It was a deadline. We had one continuous 300-log highway drive, and a cluster of twenty HPCC servers to replay it on. Eighteen of those servers were sitting idle, because replaying a continuous drive is inherently serial: the sensor state that log 16 inherits is the sensor state that log 15 produced. Split the drive naively into fifteen-log slices and node two opens on log 16 with its filter state at zero, its boresight estimate at 0.0 degrees, and no memory whatsoever of the fifteen logs that came before it.

The obvious fix, and the one we implemented first, was to overlap the slices. Hand every node one extra log of warm-up that it reads but does not score, and give it a sixteen-log chunk instead of a fifteen-log one. Node two takes log 16 as warm-up, throws its KPI output away, and keeps logs 17 through 31. The reason we were confident this would work is that the Extended Kalman Filter at the back of the chain is genuinely cheap to bootstrap: ten to twenty frames is enough for it to establish plausible position and velocity tracks. One log of overlap buys that, and for the tracker the sixteen-log scheme is genuinely sufficient.

Then we ran it, and the angle KPIs failed on every parallel node.

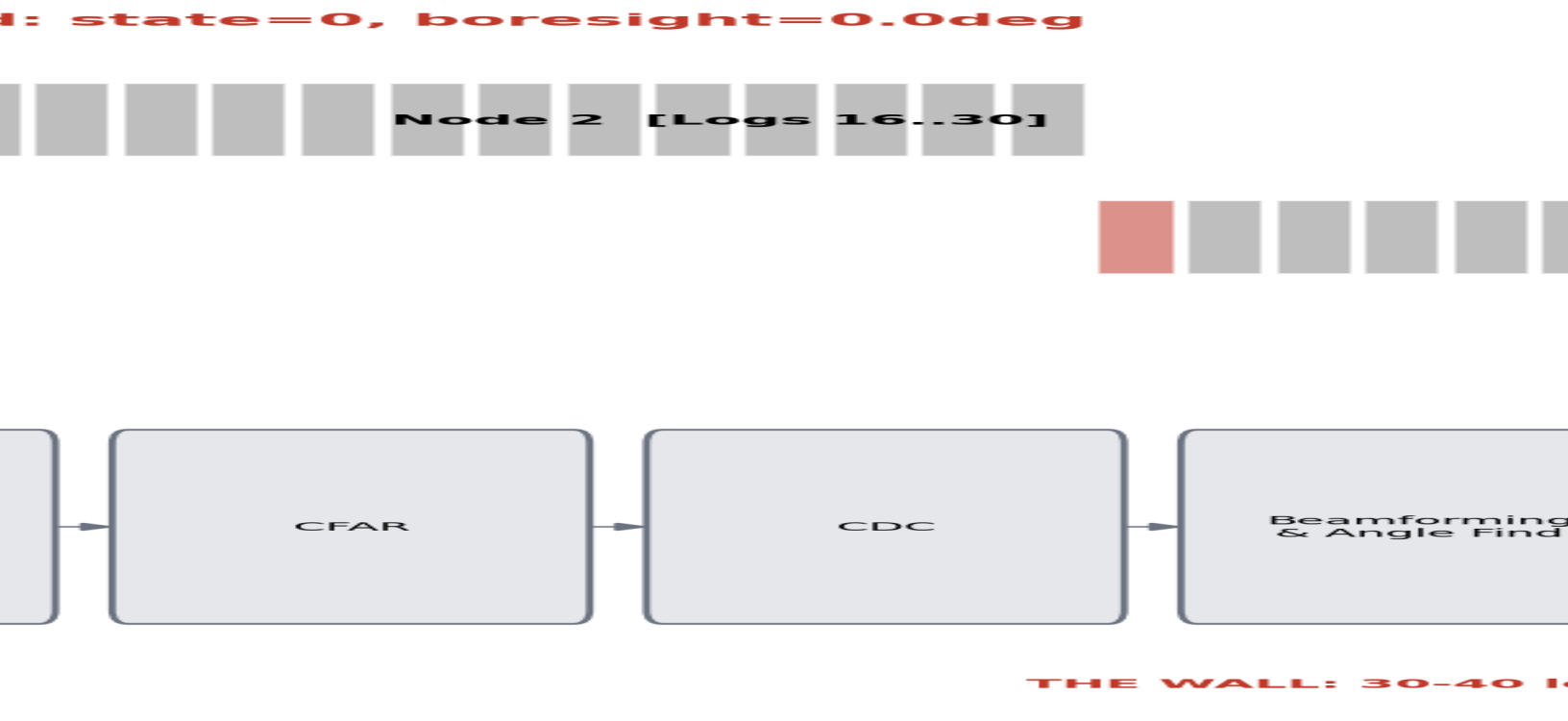



*Figure 1. (Top) Naive fifteen-log slicing: every node after the first begins cold. (Bottom) The FMCW chain and the wall. The EKF at the far right adapts within one log; AF Dynamic Alignment, immediately after beamforming, does not.*

The wall sits in an unintuitive place. It is not in the tracker, which we had already accounted for. It is in AF Dynamic Alignment, the stage that estimates the physical mounting tilt of the sensor relative to the vehicle - the small boresight bias that comes from the radar not being bolted down perfectly square. That estimate runs on a deliberately conservative static gain of K■ = 0.0025, combined with a startup deadband. It is set that way for good reason: a boresight estimate that moves fast will happily chase a guardrail or a parked truck and permanently bend the calibration. But conservative means slow, and slow means the estimator needs somewhere between thirty and forty logs, roughly four hundred frames or twenty minutes of driving, before it genuinely locks.

This is where the sixteen-log idea hit a wall of arithmetic. A thirty-log warm-up cannot be overlaid onto a fifteen-log slice. There is no arrangement of chunk boundaries that satisfies both. So every parallel node spent its entire working window inside the estimator's deadband, produced angles that were not yet trustworthy, and failed the Tier-2 KPI check. Across twenty nodes that is on the order of 360,000 frames of valid drive that simply could not be scored. The cluster did not get faster; it got slower than the single serial replay we started with.

## II. Dual-Horizon Exponential Dynamic Alignment

The insight is that the conservative gain is solving two different problems at once. In the first seconds after power-on the estimator genuinely needs to move fast, because it has no information at all. After that, it genuinely needs to move slowly, because it has good information that a fast gain would let it destroy. The static schedule cannot serve both, because it picks one number and keeps it forever. So we replaced the constant with a function that starts high and decays into the original constant, and we made the decay the same constant it decays to.

```
BEFORE K(N) = K_slow = 0.0025 for all N

requires N_wait = 400 frames (30-40 logs, ~20 min)


AFTER K_DA(N) = K_fast * exp(-alpha * N) + K_slow

subject to | v_r + v_ego * cos(theta) | <= epsilon


K_fast = 0.05 K_slow = 0.0025 alpha = 0.08

N = scan index
```

The intuition is the crooked-glasses one. When you put on glasses that are not sitting straight, you make a large, fast correction in the first second or two - that is K■■■ at 0.05, twenty times the steady-state gain. Then you settle, because you have found the position and you do not want a pothole to knock you off it - that is the exponential decay into K■lln at 0.0025, the original conservative value, left exactly where the legacy calibration expects to find it.

The second half of the equation matters just as much, and it is the part that stops this from being a liability. Fast adaptation is only safe when the candidate being tracked is plausibly stationary in the ego frame. The Doppler gate enforces exactly that: a genuine boresight bias is a geometric error, so the object generating it should have a radial velocity consistent with being bolted to the world and passing us at our own speed. A guardrail or a parked vehicle drifting into the gate does not satisfy this, and the estimator declines to move fast for it. The gain schedule and the gate were designed together; either one alone would be a regression.

In firmware this is four lines in ghost_calib.c:

```
float DaDualHorizonGain(float k_fast, float k_slow, float alpha, int32_t
n)

{

/* K_DA(n) = K_fast * exp(-a*n) + K_slow. n>=0. */

return k_fast * expf(-alpha * (float)n) + k_slow;

}
```

It is deliberately that small, and deliberately that ordinary. It runs on the BBE32 alongside the existing dynamic-alignment code, it costs one expf per scan, and it composes with the legacy Auto_align entry points in dyn_align_wrapper.c rather than replacing them - the wrapper's Auto_align_Init() simply caps the startup deadband at 50 frames now, which was a 400-frame wait that no longer serves any purpose once the gain schedule is doing the work.

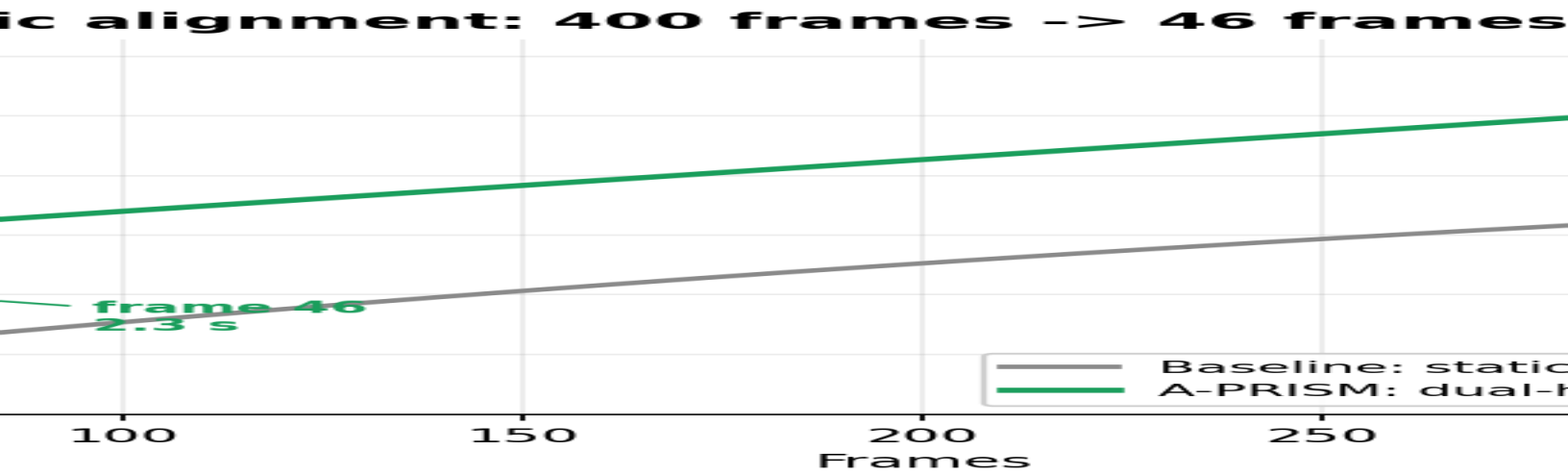



*Figure 2. Boresight azimuth error against scan count. The baseline crawls from -1.145 degrees to zero across the full 400-frame window; the dual-horizon schedule reaches the lock criterion at frame 46, which is 2.3 seconds of driving.*

## III. Preflight Seeding and Multi-Radar Temporal Deskewing

Forty-six frames is a great answer for a vehicle that has just booted. It is still the wrong answer for a cluster replaying a log that was recorded yesterday, because in that case the correct answer is zero and we already know what it is. The converged boresight angles are sitting in the log header, written by the vehicle that originally recorded the drive. So we read them.

```
seed = ( AutoAlignAzimuth , AutoAlignElevation ) read at t = 0
```

```
accepted only if |az| <= 5 deg and |el| <= 2 deg

Aprism_DC_Preflight_Seed(conv_az_deg, conv_el_deg) -> 1 = accepted
```

The validation bounds are not decoration. A header from a drive that never converged carries a garbage angle, and a garbage seed is worse than no seed at all because the estimator now believes it. Anything outside a physically plausible mounting window is refused and the replay falls back to the ordinary gain schedule, which is slower but correct. Replay nodes in our 300-log cluster now start aligned at frame 1.

Seeding fixes the cold start, but multi-sensor fusion has a second and subtler timing problem. Our satellites do not all look at the world at the same instant. Across a radar set we measure time offsets between five and forty-five milliseconds, and at highway closing speeds even a forty millisecond error is a visible smear - a car ahead appears smeared along its own path rather than at a point. We correct for it by extrapolating each satellite's measurement back onto a common ego-frame timestamp, which on the SE(2) manifold is a rotation followed by a translation:

```
p_corr = R_z(omega * dt) * p_meas - v_ego * dt - 1/2 * a_ego * dt^2

dt in [5, 45] ms measured effect: 0.68 m -> 0.09 m (-86.2%)
```

Separately, a clock bias and a velocity bias are not distinguishable from a single radar's own measurements - both look like a translation that drifts with speed. We break the degeneracy with two simultaneously tracked targets, using a Gram-type linear invariant in the range measurements that removes yaw from the problem entirely:

```
r_k * w_k - d_k^T * v_k = || v_k ||^2 * tau

two targets, no yaw term timing RMSE: 3.8 ms
```

What remains is relative yaw between satellites, which cancels in the difference D■■ = e^(-jβ■)(p■ - p■*). Taken together these three steps are what make a twenty-node replay produce numbers that match the original drive rather than merely plausible ones.

## IV. Low-Compute DSP Refinement

The last piece is on the sensor and is a resource problem rather than an accuracy problem. Tightening a target's range means estimating the residual Doppler between what we predicted and what actually came back. The brute-force way is a matched filter across the whole lag bank, and on the BBE32 that is 4,608 multiplications per target per scan. We do not need all of them: derotate the fast-time samples by the predicted frequency first, and the residual collapses into a lag-1 autocorrelation.

```
z[n] = x[n] * exp(-j*2*pi*f_pred*n*T_s)

f_res = angle( sum z[n] * conj(z[n-1]) ) / ( 2*pi*T_s )
```

```
4,608 -> 1,024 multiplications (4.5x fewer) INT8 range RMSE: 0.108 m
```

Conditioning the estimator on an already-confirmed track means we only pay it where there is a track worth refining, which is most of the saving. The implementation lives in residual_track_ifc.c and is gated on a track being present, so it costs nothing on an empty road.

## V. Verification

Every module described here sits behind a compile guard that defaults to off, which means the off build must be bit-identical to the shipped baseline. That is not a style preference; it is the only reason the change is mergeable into a safety-relevant firmware branch. We built both configurations and ran the same 40-test host harness against each.

| Build | Configuration | Result | Meaning |
|---|---|---|---|
| aprism_ab_off.exe | APRISM_*_ENABLE = 0 | 40 / 40 PASS | Baseline path is unchanged |
| aprism_ab_on.exe | APRISM_*_ENABLE = 1 | 40 / 40 PASS | Feature path is sound |
| MSVC cl /std:c++17 | both configs, standalone headers | clean compile | No hidden toolchain dependency |

The module-level A/B benchmark isolates the behaviour change. T1 confirms the default fill is untouched (0.5 both sides). T2 exercises the spray path and shows the covariance inflation engaging as designed. T3 and T4 confirm the sentinel contract described in our companion paper: a ghost that has been assimilated reads -1 rather than being penalised or, in the behind-guardrail case, protected by accident.

The cluster-level result is the one that motivated the work. Across a 300-log drive replayed on twenty nodes, the baseline scheme could not keep any node's angle KPIs within tolerance during its working window; the dual-horizon schedule brings every node inside tolerance within 46 frames of its first scan, and preflight seeding brings replayed nodes inside tolerance at frame 1.

# Appendix A — Invention Disclosure (IDF) & Patent Claims

The following claims are drafted for internal invention disclosure. Claim 1 is the independent system claim; claims 2-9 are dependent. The commercial-detectability notes identify which observable behaviours a competitor product would exhibit, so the claims are anchored to measurable outputs rather than implementation detail.

## *Claim 1 (independent — system)*

A method for reducing cold-start convergence latency in a sensor boresight alignment estimator of an automotive radar, comprising: maintaining an alignment gain that is the sum of a fast gain term attenuated by an exponential decay in scan index and a constant steady-state gain term; applying the gain to update an azimuth and/or elevation boresight estimate; and suppressing acceleration of the gain update when a radial velocity of a candidate reflector, corrected by an ego-vehicle velocity component projected onto the candidate bearing, exceeds a predetermined threshold, such that the fast gain term is applied only while the candidate is consistent with a stationary object in the ego frame.

**Claim 2.** The method of claim 1, wherein the fast gain term is approximately 0.05 and the steady-state gain term is approximately 0.0025, and the exponential decay constant is approximately 0.08 per scan.

**Claim 3.** The method of claim 1, further comprising storing, in a header of a recorded sensor log, the converged boresight azimuth and elevation at end of a driving session, and on replay seeding the alignment estimator from the stored values at a first scan of the replay.

**Claim 4.** The method of claim 3, further comprising accepting the seed only when the stored azimuth and elevation lie within predetermined physical bounds, and otherwise falling back to the gain schedule of claim 1.

**Claim 5.** The method of claim 1, further comprising extrapolating measurements from a plurality of satellite sensors onto a common ego-frame timestamp by a rotation through an ego yaw rate integrated over an inter-sensor time offset, followed by translation by an ego velocity and one-half an ego acceleration.

**Claim 6.** The method of claim 5, further comprising estimating a clock offset from a linear invariant formed from range measurements to at least two concurrently tracked targets, the invariant being independent of relative satellite yaw.

**Claim 7.** The method of claim 1, further comprising estimating a residual Doppler for a confirmed track by derotating fast-time samples by a predicted frequency and evaluating a lag-1 autocorrelation of the derotated samples.

**Claim 8.** The method of claim 7, wherein the residual Doppler estimation is executed only for targets already confirmed by a tracker, thereby reducing multiply-accumulate operations by at least a factor of four.

**Claim 9.** The method of claim 1, wherein the fast gain term brings the boresight error within a lock tolerance in fewer than one hundred scans from a cold start,

## *Commercial detectability criteria*

| Claim | Observable in a competitor product | Detection method |
|---|---|---|
| 1 | Cold-start angle error persists for many minutes of driving after module reset | Replay a public drive log with the estimator reset; measure azimuth error vs scan index |
| 2 | Gain schedule shape distinguishable from a pure constant | Infer effective gain from the exponential decay of error over the first 100 scans |
| 3 | Boresight angles present in a log header | Parse recorded log headers for alignment fields populated before the driving session begins |

| Claim | Observable in a competitor product | Detection method |
|---|---|---|
| 4 | Seed rejection thresholds observable | Force a header with implausible angles; observe whether the estimator locks or refuses |
| 5 | Multi-sensor fusion smear proportional to inter-sensor timing skew | Measure apparent target extent along its own velocity trace at high closing speed |
| 6 | Clock-bias estimator present | Perturb one satellite's timestamp in software; observe whether the bias estimate corrects it |
| 7-8 | Reduced DSP multiply count | Profile an SoC during track refinement; look for a lag-1 autocorrelation kernel rather than a full lag bank |

Novelty position: the exponential two-horizon gain schedule, the Doppler-gated fast/slow arbitration, and the use of a recorded converged header as a replay seed are, to our knowledge, not taught in the radar calibration literature, which treats alignment gain as a fixed design constant or as a pure learning-rate schedule with no physical admission test attached.